Question Bank

MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

Machine Learning for Data Science

III B.Tech I Semester

(2025-26)

Department of CSE (Data Science)

School of Engineering

School of Engineering-A.Y 2025-2026 Regulation: R23

(CSE-DATASCIENCE)

III Year B. Tech - I Semester

Subject Name: MACHINE LEARNING FOR DATA SCIENCE

Subject Code: MR23-1CS0305

QUESTION BANK

Qno	Question	Marks	Section
1	Define the concept of Machine Learning and outline its major applications across different domains.	8	**Section-I**
2	Compare Deep Learning with traditional Machine Learning, and illustrate key real-world applications of Deep Learning.	8	**Section-I**
3	Classify and describe different types of Machine Learning with suitable examples.	8	**Section-I**
4	What are the main steps involved in the process of Exploratory Data Analysis (EDA)?	8	**Section-I**
5	Discuss various statistical and visualization-based techniques used to detect outliers in datasets.	8	**Section-I**
6	Evaluate different strategies for handling detected outliers and justify the most suitable one for a given scenario.	8	**Section-I**
7	a) Explain the concept of Feature Engineering in Machine Learning. b) Demonstrate and discuss common techniques used to create effective features.	4 4	**Section-I**
8	Illustrate the process of Principal Component Analysis (PCA) and analyse how it reduces dimensionality while retaining key information.	8	**Section-I**
9	Differentiate among Filter, Wrapper, and Embedded methods for Feature Selection with examples.	8	**Section-I**
10	Explain the following and their purpose. Min-Max Normalization, Z-Score Normalization, Robust Scaling	8	**Section-I**
11	Define Regression and explain the concept of Logistic Regression with suitable example.	8	**Section-II**
12	a) Write a Python program to implement a Simple Linear Regression model. b) Write an interpretation of the results obtained from the model implementation.	4 4	**Section-II**
13	Implement a Multiple Linear Regression model using Python and explain the outcomes.	8	**Section-II**
14	Analyse the need for Adjusted R-Squared over R-Squared in model evaluation and interpret their implications.	8	**Section-II**
15	Explain R-Squared and demonstrate how it represents the goodness of fit in a regression model.	8	**Section-II**
16	Describe the Gradient Descent algorithm and analyse how it minimizes the cost function in regression models.	8	**Section-II**
17	Define the cost function and write a Python program to perform the following: a) Load data from features.csv and target.csv.	4 4	**Section-II**

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	b) Train a Simple Linear Regression model and Compute and print MAE, MSE, R^2, and Adjusted R^2.		
18	Describe the following Evaluation Metrics: a) Mean Squared Error (MSE) b) Mean Absolute Error (MAE)	4 4	**Section-II**
19	Justify the use of Logistic Regression instead of Linear Regression in certain predictive modelling scenarios.	8	**Section-II**
20	List and explain the key assumptions made by Linear Regression models.	8	**Section-II**
21	Explain the working mechanism of Decision Trees with a suitable example.	8	**Section-III**
22	Analyse how entropy and information gain are used for node splitting in Decision Trees.	8	**Section-III**
23	Describe the K-Nearest Neighbors (KNN) algorithm. How does the choice of 'K' affect the performance of the model?	8	**Section-III**
24	Explain the functioning of Support Vector Machines (SVM) and analyse the role of hyperplanes and support vectors.	8	**Section-III**
25	a) Write an explanation of the *Precision–Recall trade-off* in classification models. b) Evaluate the *impact of this trade-off* on overall model performance.	4 4	**Section-III**
26	Construct and explain a Confusion Matrix, describing its four major components.	8	**Section-III**
27	Examine the bias-variance trade-off in Decision Trees and Random Forests and justify how Random Forest mitigates overfitting.	8	**Section-III**
28	Explain how Random Forest improves performance over a single Decision Tree model.	8	**Section-III**
29	Describe the Naive Bayes algorithm and identify appropriate applications for its use.	8	**Section-III**
30	Interpret ROC curves and AUC values to compare classification model performance.	8	**Section-III**
31	Define Ensemble Methods and explain their role in improving model performance.	8	**Section-IV**
32	Differentiate between Bagging and Boosting techniques with examples.	8	**Section-IV**
33	Explain the concept and working of the Stacking Ensemble Technique.	8	**Section-IV**
34	Describe Voting in Ensemble Learning and illustrate its implementation.	8	**Section-IV**
35	**a)** Write a short note explaining the *Holdout Method* for model validation. **b)** Write an evaluation discussing the *effectiveness and limitations* of	4 4	**Section-IV**

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	the Holdout Method.		
36	Describe the K-Fold Cross-Validation process and analyse its advantages over other validation methods.	8	**Section-IV**
37	Explain the Bias-Variance Trade-off and assess its effect on machine learning model performance.	8	**Section-IV**
38	**a)** Discuss various *Regularization techniques* used in Machine Learning. **b)** Compare *L1* and *L2 Regularization* with suitable examples.	4 4	**Section-IV**
39	Differentiate between Overfitting and Underfitting with appropriate illustrations.	8	**Section-IV**
40	Evaluate the effectiveness of Bagging and Boosting in minimizing model overfitting.	8	**Section-IV**
41	Define the goal of clustering in unsupervised learning and explain its applications.	8	**Section-V**
42	Explain the working steps of the K-Means algorithm and analyse its limitations.	8	**Section-V**
43	Describe Hierarchical Clustering and interpret the meaning of a dendrogram.	8	**Section-V**
44	Explain DBSCAN and evaluate how it differs from K-Means clustering.	8	**Section-V**
45	Compare Gaussian Mixture Models (GMMs) and K-Means in handling cluster shapes and data distribution.	8	**Section-V**
46	a) Write a description of the Expectation (E-step) and Maximization (M-step) in the EM algorithm. b) Write an example to illustrate how the E-step and M-step work in practice.	4 4	**Section-V**
47	Analyse how the EM algorithm manages missing or incomplete data during clustering.	8	**Section-V**
48	Explain why feature scaling is crucial for clustering algorithms like K-Means.	8	**Section-V**
49	a) Compare K-Means, DBSCAN, and GMMs in terms of their approach, advantages, and limitations. b) Recommend suitable scenarios or applications where each clustering algorithm is most effective.	4 4	**Section-V**
50	Explain the Elbow Method to determine the optimal number of clusters in K-Means and demonstrate how to identify the 'elbow' point.	8	**Section-V**